

Every Physics Formula

Class XII CBSE · Physics · Chapters 1–9

Every Physics Formula

Nine chapters, one hundred entries. Every one carries what its symbols mean and the SI unit it comes out in, because in physics a formula without its units is half a formula. The cue is on the outside and the formula is hidden, so you can read down a chapter and test yourself rather than just re-reading.

- Entries: 100
- Chapters: 9
- Must be instant: 78
- Constants: 14

How to use this

● means it should arrive before you have finished reading the question. ○ means a few seconds is fine — these are the ones you reconstruct from a derivation rather than recall outright, and the companion page **Physics, Derived** shows how.

Each chapter opens with its own **recognise strip**: every cue in that chapter on one screen. Read the strip cold, name the formula, then tap the entry to check. That drill is worth more than re-reading the page.

Vector arrows are kept where the direction matters and dropped where only magnitude is asked. Where a symbol is overloaded across chapters — L is length, inductance and angular momentum in three different chapters — the entry says which one it means.

CH 1 Electric Charges and Fields — 13 entries

Recognise strip — say it, then open the entry

- P1.1 Charge is quantised, and conserved
- P1.2 Force between two point charges
- P1.3 The same force with a medium in the gap
- P1.4 Force from several charges at once
- P1.5 Field of a point charge, and what field means
- P1.6 Charge spread along a line, over a surface, through a volume
- P1.7 Dipole moment
- P1.8 Field on the axis of a dipole
- P1.9 Field on the equatorial line of a dipole
- P1.10 Torque and energy of a dipole in a uniform field

- P1.11 Electric flux, and Gauss's law
- P1.12 Field of an infinite line charge and of a plane sheet
- P1.13 Field of a charged spherical shell, inside and out

● P1.1 Charge is quantised, and conserved

$$q = \pm ne \quad e = 1.6 \times 10^{-19} \text{ C}$$

Symbol	Meaning	Unit
q	charge on a body	C
n	an integer — never a fraction	dimensionless
e	elementary charge	$1.6 \times 10^{-19} \text{ C}$

Use it when:

Asked how many electrons make up a given charge, or whether a stated charge is possible. 1 C is about 6.25×10^{18} elementary charges — which is why the coulomb is an impractically large unit.

Trap:

Charge is also **conserved** and **additive**: it is a scalar, so charges add algebraically with their signs, never as vectors.

● P1.2 Force between two point charges

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} = k \frac{Q_1 Q_2}{r^2} \quad \vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r_{12}^2} \hat{r}_{12}$$

Symbol	Meaning	Unit
F	force between the charges	N
k	$1/4\pi\epsilon_0$	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
ϵ_0	permittivity of free space	$8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
r	separation	m

Use it when:

Two charges, and a force. Coulomb's law. Valid only for **point charges** — separation much larger than the bodies themselves.

Trap:

Unlike gravitation, this force **depends on the medium** and can be attractive or repulsive. $\vec{F}_{12} = -\vec{F}_{21}$ by Newton's third law even when the charges are unequal.

○ P1.3 The same force with a medium in the gap

$$F = \frac{1}{4\pi\epsilon_0 K} \frac{Q_1 Q_2}{r^2}, \quad K = \epsilon_r = \frac{\epsilon}{\epsilon_0}$$

$$\text{partly filled: } F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{\left[(r-t) + \sqrt{K}t\right]^2}$$

Symbol	Meaning	Unit
K	dielectric constant (relative permittivity)	dimensionless
t	thickness of the dielectric slab	m
ϵ	absolute permittivity of the medium	C ² N ⁻¹ m ⁻²

Use it when:

The charges sit in water, oil or a slab rather than vacuum. A dielectric always **reduces** the force, since $K > 1$.

Trap:

A slab of thickness t behaves like $\sqrt{K}t$ of vacuum — the square root is easy to lose.

● P1.4 Force from several charges at once

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \dots$$

Symbol	Meaning	Unit
$\vec{F}_{1j}$	force on charge 1 from charge j alone	N
$\vec{F}_1$	resultant, by vector addition	N

Use it when:

Three or more charges. Principle of superposition: compute each pair as if the others were absent, then add as vectors.

Trap:

Add as **vectors**, not magnitudes. For a null-point question, first decide whether the point lies between the charges (same signs) or outside, beyond the weaker one (opposite signs) — then solve.

● P1.5 Field of a point charge, and what field means

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0} \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \quad \vec{F} = q\vec{E}$$

Symbol	Meaning	Unit
$\vec{E}$	electric field intensity	N C ⁻¹ or V m ⁻¹
q_0	test charge, taken vanishingly small	C
Q	source charge	C

Use it when:

Anything about field strength at a point. E falls as $1/r^2$; to straighten the graph, plot E against $1/r^2$.

Trap:

The test charge cancels — E is a property of the source and the point, not of what you probe it with. The limit exists so the probe does not disturb the field it measures.

○ P1.6 Charge spread along a line, over a surface, through a volume

$$\lambda = \frac{dq}{dl} \quad \sigma = \frac{dq}{dS} \quad \rho = \frac{dq}{dV}$$

Symbol	Meaning	Unit
λ	linear charge density	C m ⁻¹
σ	surface charge density	C m ⁻²
ρ	volume charge density	C m ⁻³

Use it when:

The charge is continuous rather than a set of points — a charged wire, sheet or sphere. These are what turn a sum into an integral, and they feed straight into the Gauss's law applications.

Trap:

On a **conductor** the charge sits entirely on the outer surface, so σ is the relevant density even for a solid sphere.

● P1.7 Dipole moment

$$\vec{p} = Q \times 2\vec{l}$$

Symbol	Meaning	Unit
$\vec{p}$	electric dipole moment	C m
Q	magnitude of either charge	C
$2l$	separation between the charges	m

Use it when:

Any dipole question. Direction is **from the negative charge to the positive one** — the opposite of the field it produces between them.

Trap:

The separation is $2l$, not l . Losing the 2 halves every subsequent answer.

● **P1.8 Field on the axis of a dipole**

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - l^2)^2} \xrightarrow{r \gg l} \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

Symbol	Meaning	Unit
E_{axial}	field on the dipole's axis	N C^{-1}
r	distance from the dipole's centre	m
p	dipole moment	C m

Use it when:

The point lies on the line through both charges. Direction is **parallel** to $\vec{p}$.

Trap:

$1/r^3$, not $1/r^2$ — a dipole's field dies faster than a point charge's because the two charges nearly cancel.

● **P1.9 Field on the equatorial line of a dipole**

$$E_{\text{eq}} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + l^2)^{3/2}} \xrightarrow{r \gg l} \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$$

Symbol	Meaning	Unit
E_{eq}	field on the perpendicular bisector	N C^{-1}
r	distance from the centre	m

Use it when:

The point is on the perpendicular bisector of the dipole.

Trap:

Exactly **half** the axial field at the same distance, and pointing **antiparallel** to $\vec{p}$. Both the factor of two and the direction are asked.

● **P1.10 Torque and energy of a dipole in a uniform field**

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta \quad U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

$$W = pE (\cos \theta_1 - \cos \theta_2)$$

Symbol	Meaning	Unit
τ	torque	N m
U	potential energy of the dipole	J
θ	angle between $\vec{p}$ and $\vec{E}$	rad or °

Use it when:

A dipole is placed in a field and asked to rotate. Net **force** in a uniform field is zero — only a torque acts.

Trap:

$\theta = 0$ is stable equilibrium ($U = -pE$, minimum); $\theta = 180^\circ$ is unstable ($U = +pE$). Energy is measured from $\theta = 90^\circ$, where $U = 0$.

● P1.11 Electric flux, and Gauss's law

$$\Phi = \vec{E} \cdot \vec{A} = EA \cos \theta \quad \oint_S \vec{E} \cdot d\vec{S} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Symbol	Meaning	Unit
Φ	electric flux	N m ² C ⁻¹ or V m
$\vec{A}$	area vector, normal to the surface	m ²
Q_{enc}	net charge inside the closed surface	C

Use it when:

Symmetry lets you pick a Gaussian surface on which E is constant — a sphere, a cylinder, a pillbox.

Trap:

Only **enclosed** charge counts. A charge outside contributes zero net flux (what enters, leaves) — but it still contributes to $\vec{E}$ at every point on the surface.

● P1.12 Field of an infinite line charge and of a plane sheet

$$E_{\text{line}} = \frac{\lambda}{2\pi\epsilon_0 r} \quad E_{\text{sheet}} = \frac{\sigma}{2\epsilon_0} \quad E_{\text{conductor surface}} = \frac{\sigma}{\epsilon_0}$$

Symbol	Meaning	Unit
λ	linear charge density	C m ⁻¹
σ	surface charge density	C m ⁻²
r	perpendicular distance from the wire	m

Use it when:

A long charged wire, or a large charged plate. Both come straight out of Gauss's law.

Trap:

The sheet field is **independent of distance** — and it is $\sigma/2\epsilon_0$ for a thin sheet with field on both sides, but σ/ϵ_0 just outside a **conductor**, where the field exists on one side only. Confusing the two is the classic error.

● P1.13 Field of a charged spherical shell, inside and out

$$r > R : E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad r = R : E = \frac{\sigma}{\epsilon_0} \quad r < R : E = 0$$

Symbol	Meaning	Unit
R	radius of the shell	m
r	distance from the centre	m
q	total charge on the shell	C

Use it when:

A hollow charged sphere or any charged conductor. Outside, it behaves exactly as if all the charge sat at the centre.

Trap:

$E = 0$ inside but the **potential is not zero** — it is constant at the surface value. Zero field means no *change* in potential, not no potential.

CH 2 Electrostatic Potential and Capacitance — 14 entries

Recognise strip

- P2.1 What potential and potential difference mean
- P2.2 Potential from the field, and the field back from potential
- P2.3 Potential of a point charge and of a system
- P2.4 Potential due to a dipole at any point
- P2.5 Potential of a charged spherical shell, inside and out
- P2.6 Energy of a system of point charges

- P2.7 Energy of a charge, and of a dipole, in an external field
- P2.8 Capacitance, defined
- P2.9 Parallel plate capacitor, with and without a dielectric
- P2.10 A dielectric slab only partly filling the gap
- P2.11 Spherical and cylindrical capacitors
- P2.12 Capacitors in series and in parallel
- P2.13 Energy stored in a capacitor, and energy density
- P2.14 Two charged capacitors joined together

● P2.1 What potential and potential difference mean

$$V_A - V_B = \frac{W_{B \rightarrow A}}{q_0} \quad V = \frac{W_{\infty \rightarrow P}}{q_0}$$

Symbol	Meaning	Unit
V	electric potential	V (= J C ⁻¹)
W	work done by an external force, without acceleration	J
q_0	test charge moved	C

Use it when:

Work, energy or volts are involved. Potential is a **scalar**, which is what makes it easier to work with than $\vec{E}$.

Trap:

"Without acceleration" means the external force exactly balances the electric force, so no kinetic energy is gained. Convention: $V(\infty) = 0$.

● P2.2 Potential from the field, and the field back from potential

$$V = - \int_B^A \vec{E} \cdot d\vec{l} \quad E = - \frac{dV}{dr} \quad \oint \vec{E} \cdot d\vec{l} = 0$$

Symbol	Meaning	Unit
E	field component along r	V m ⁻¹
dV/dr	potential gradient	V m ⁻¹

Use it when:

Converting between E and V , or asked which of two points is at higher potential.

Trap:

The minus sign says $\vec{E}$ points toward **decreasing** potential. The closed-loop integral being zero is the statement that the electrostatic field is **conservative**.

● P2.3 Potential of a point charge, and of a system

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad V_P = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

Symbol	Meaning	Unit
V	potential at distance r	V
q_i	each charge, with its sign	C
r_i	distance from each charge to the point	m

Use it when:

Potential at a point from one or several charges. Falls as $1/r$, unlike E 's $1/r^2$.

Trap:

A **plain algebraic sum** — no vectors. Midway between $+q$ and $-q$, $V = 0$ but $\vec{E} \neq 0$. $V = 0$ never implies $E = 0$.

○ P2.4 Potential due to a dipole at any point

$$V = \frac{p \cos \theta}{4\pi\epsilon_0 (r^2 - l^2 \cos^2 \theta)} \xrightarrow{r \gg l} \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$$

Symbol	Meaning	Unit
θ	angle from the dipole axis	rad or °
p	dipole moment	C m
r	distance from the dipole centre	m

Use it when:

A general point, not just axial or equatorial. Axial is $\theta = 0$; equatorial is $\theta = 90^\circ$.

Trap:

$V = 0$ everywhere on the **equatorial line** ($\cos 90^\circ = 0$), even though $\vec{E}$ is not zero there. Potential of a dipole falls as $1/r^2$, faster than a point charge's $1/r$.

● P2.5 Potential of a charged spherical shell, inside and out

$$r \geq R: V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad r < R: V = V_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

Symbol	Meaning	Unit
R	radius of the shell	m
V	potential	V

Use it when:

Any charged conductor. Sketching V against r is a standard question: flat inside, then $1/r$ outside.

Trap:

Constant inside, not zero — even though $E = 0$ there. This pairs with P1.13 and the two are examined together.

● P2.6 Energy of a system of point charges

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} \quad U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

Symbol	Meaning	Unit
U	potential energy of the assembled system	J
r_{ij}	separation of each pair	m

Use it when:

"Work done in assembling" a set of charges. Sum over **every distinct pair**, once each.

Trap:

Three charges give three pairs, not three terms of one charge each. Keep the signs — a mixed set can give negative U .

○ P2.7 Energy of a charge, and of a dipole, in an external field

$$U = qV(\vec{r}) \quad U_{\text{dipole}} = -\vec{p} \cdot \vec{E} \quad 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$U_{\text{two charges in a field}} = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Symbol	Meaning	Unit
$V(\vec{r})$	potential of the external field at that point	V
U	potential energy	J or eV

Use it when:

Charges are placed in a field produced by something else. The last term is the charges' mutual energy — easy to forget.

Trap:

An electron-volt is an energy, not a voltage: the energy one elementary charge gains falling through 1 V.

● **P2.8 Capacitance, defined**

$$C = \frac{Q}{V} \quad C_{\text{isolated sphere}} = 4\pi\epsilon_0 R$$

Symbol	Meaning	Unit
C	capacitance	F (farad)
Q	charge on either plate	C
V	potential difference across it	V

Use it when:

Any capacitor question. C depends only on geometry and the dielectric — never on Q or V .

Trap:

The farad is enormous; real capacitors are μF , nF or pF . Q is the charge on **one** plate, the two being equal and opposite.

● **P2.9 Parallel plate capacitor, with and without a dielectric**

$$C = \frac{\epsilon_0 A}{d} \quad C_{\text{with dielectric}} = \frac{K\epsilon_0 A}{d}$$

Symbol	Meaning	Unit
A	area of one plate	m^2
d	plate separation	m
K	dielectric constant	dimensionless

Use it when:

The standard capacitor. Inserting a dielectric multiplies C by K .

Trap:

What stays fixed depends on the circuit. **Battery connected** $\rightarrow V$ fixed, so Q rises. **Battery disconnected** $\rightarrow Q$ fixed, so V falls. Almost every dielectric question turns on this distinction.

○ **P2.10 A dielectric slab only partly filling the gap**

$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{K}} \quad \text{several slabs: } C = \frac{\epsilon_0 A}{\frac{t_1}{K_1} + \frac{t_2}{K_2} + \dots}$$

Symbol	Meaning	Unit
t	thickness of the slab	m
d	plate separation	m
K	dielectric constant of the slab	dimensionless

Use it when:

A slab thinner than the gap is slid in. Setting $t = d$ recovers $K\varepsilon_0 A/d$; setting $K = 1$ recovers $\varepsilon_0 A/d$ — use those two checks.

Trap:

The result does not depend on **where** in the gap the slab sits.

○ P2.11 Spherical and cylindrical capacitors

$$C_{\text{spherical}} = \frac{4\pi\varepsilon_0 r_1 r_2}{r_2 - r_1} \quad C_{\text{cylindrical}} = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}$$

Symbol	Meaning	Unit
r_1, r_2	inner and outer radii	m
a, b	inner and outer cylinder radii	m
L	length of the cylinder	m

Use it when:

Concentric shells or coaxial cylinders. Both are derived by integrating E between the conductors to get V , then $C = Q/V$.

Trap:

Here L is a **length**. In Chapter 6 the same letter is self-inductance — check which chapter the question is in.

● P2.12 Capacitors in series and in parallel

$$\text{series: } \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \dots \quad \text{parallel: } C_p = C_1 + C_2 + \dots$$

Symbol	Meaning	Unit
series	same charge , voltages add	F
parallel	same voltage , charges add	F

Use it when:

A network of capacitors. Reduce it stepwise.

Trap:

The rules are the **opposite way round** from resistors. Series capacitance is always smaller than the smallest one in the chain.

● **P2.13 Energy stored in a capacitor, and energy density**

$$U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C} \quad u = \frac{1}{2}\epsilon_0 E^2$$

Symbol	Meaning	Unit
U	energy stored	J
u	energy per unit volume of field	J m^{-3}
E	field between the plates	V m^{-1}

Use it when:

Energy, or work done in charging. Pick whichever of the three forms matches what the question gives you.

Trap:

The factor of $\frac{1}{2}$ is because V rises from 0 to its final value as charge accumulates — the work is not simply QV .

○ **P2.14 Two charged capacitors joined together**

$$V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \quad \Delta U = \frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$$

Symbol	Meaning	Unit
V_{common}	shared potential after connecting	V
ΔU	energy lost in the process	J

Use it when:

Two charged capacitors are connected plate to plate. Charge is conserved; energy is not.

Trap:

Energy is **always lost** (the expression is a square), dissipated as heat and radiation in the connecting wires — even ideal ones. Being asked to explain that loss is common.

CH 3 Current Electricity — 14 entries

Recognise strip

- P3.1 Current, and current density
- P3.2 Drift velocity, and the current it produces
- P3.3 Mobility
- P3.4 Ohm's law, and resistance from dimensions
- P3.5 Resistivity from what the electrons are doing
- P3.6 Conductivity, and Ohm's law in microscopic form
- P3.7 Resistance changing with temperature
- P3.8 Resistors in series and parallel
- P3.9 Emf, terminal voltage and internal resistance
- P3.10 Cells combined in series and in parallel
- P3.11 Electrical power and energy
- P3.12 Kirchhoff's two rules
- P3.13 Wheatstone bridge, and the metre bridge
- P3.14 Potentiometer — comparing emfs and finding internal resistance

● P3.1 Current, and current density

$$I = \frac{Q}{t} = \frac{dQ}{dt} \quad \vec{J} = \frac{I}{A} \hat{n}, \quad I = \int_A \vec{J} \cdot d\vec{A}$$

Symbol	Meaning	Unit
I	current — a scalar	A
$\vec{J}$	current density — a vector	A m ⁻²
A	cross-sectional area	m ²

Use it when:

Charge flow. 1 A is one coulomb per second.

Trap:

Current is drawn with an arrow but is **not a vector** — it does not obey vector addition. Bending a wire does not change the current through it. Current density is the vector.

● P3.2 Drift velocity, and the current it produces

$$\vec{v}_d = -\frac{e\vec{E}}{m}\tau \quad I = neAv_d$$

Symbol	Meaning	Unit
v_d	drift velocity	m s^{-1}
n	free electron number density	m^{-3}
τ	relaxation time between collisions	s
e	electronic charge	$1.6 \times 10^{-19} \text{ C}$

Use it when:

Linking the microscopic picture to the measured current.

Trap:

Drift velocity is tiny — around 10^{-4} m s^{-1} — while random thermal speed is about 10^5 m s^{-1} . The lamp lights instantly because the **field** propagates at nearly c , not the electrons.

○ **P3.3 Mobility**

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

Symbol	Meaning	Unit
μ	mobility	$\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$
m	mass of the charge carrier	kg
τ	relaxation time	s

Use it when:

Drift speed per unit field is asked, or comparing carriers in a semiconductor.

Trap:

Defined as a positive magnitude even for electrons, which drift opposite to $\vec{E}$.

● **P3.4 Ohm's law, and resistance from dimensions**

$$V = IR \quad R = \frac{\rho l}{A}$$

Symbol	Meaning	Unit
R	resistance	Ω
ρ	resistivity — a material property	$\Omega \text{ m}$
l	length of the conductor	m
A	cross-sectional area	m^2

Use it when:

Almost every circuit question. Stretching a wire keeps its **volume** constant, so doubling the length quarters the area and quadruples R .

Trap:

R depends on the specimen's shape; ρ does not. Ohm's law holds only for ohmic conductors at constant temperature.

● **P3.5 Resistivity from what the electrons are doing**

$$\rho = \frac{m}{ne^2\tau}$$

Symbol	Meaning	Unit
ρ	resistivity	$\Omega \text{ m}$
n	free electron density	m^{-3}
τ	relaxation time	s
m	electron mass	$9.1 \times 10^{-31} \text{ kg}$

Use it when:

Explaining *why* resistance changes — the standard 3-mark derivation, and the basis of every temperature-dependence answer.

Trap:

Heating a **metal** shortens τ , so ρ rises. Heating a **semiconductor** raises n far more than it cuts τ , so ρ falls. Same formula, opposite behaviour.

○ **P3.6 Conductivity, and Ohm's law in microscopic form**

$$\sigma = \frac{1}{\rho} = \frac{ne^2\tau}{m} \quad \vec{J} = \sigma \vec{E}$$

Symbol	Meaning	Unit
σ	conductivity	S m^{-1}
$\vec{J}$	current density	A m^{-2}
$\vec{E}$	field inside the conductor	V m^{-1}

Use it when:

Asked for Ohm's law in vector form, or to relate J and E without mentioning a circuit.

Trap:

This σ is conductivity. In Chapter 1 the same letter is surface charge density.

● P3.7 Resistance changing with temperature

$$R_t = R_0 (1 + \alpha \Delta T) \quad \alpha = \frac{R_t - R_0}{R_0 \Delta T}$$

Symbol	Meaning	Unit
α	temperature coefficient of resistance	K^{-1} or $^{\circ}\text{C}^{-1}$
R_0	resistance at the reference temperature	Ω
ΔT	temperature rise	K

Use it when:

A resistance is quoted at two temperatures.

Trap:

α is **positive** for metals, **negative** for semiconductors and insulators, and nearly zero for alloys like nichrome and manganin — which is exactly why they are used for standard resistors.

● P3.8 Resistors in series and parallel

$$R_s = R_1 + R_2 + \dots \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Symbol	Meaning	Unit
series	same current , voltages add	Ω
parallel	same voltage , currents add	Ω

Use it when:

Reducing any network. Two in parallel: $R_p = R_1 R_2 / (R_1 + R_2)$.

Trap:

The opposite of the capacitor rules. Parallel resistance is always **less** than the smallest resistor present.

● P3.9 Emf, terminal voltage and internal resistance

$$\mathcal{E} = V + Ir \quad V = \mathcal{E} - Ir \quad I = \frac{\mathcal{E}}{R + r}$$

Symbol	Meaning	Unit
$\mathcal{E}$	emf of the cell	V
V	terminal potential difference	V
r	internal resistance	Ω

Use it when:

A real cell is driving a circuit.

Trap:

$V < \mathcal{E}$ while discharging, $V = \mathcal{E}$ only in open circuit, and $V > \mathcal{E}$ while the cell is being **charged** (current reversed).

○ P3.10 Cells combined in series and in parallel

$$\text{series: } I = \frac{N\mathcal{E}}{R + Nr} \quad \text{parallel: } I = \frac{\mathcal{E}}{R + r/N}$$

$$\text{two unequal cells: } \mathcal{E}_{eq} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2}, \quad r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

Symbol	Meaning	Unit
N	number of identical cells	dimensionless
$\mathcal{E}_{eq}$	equivalent emf	V
r_{eq}	equivalent internal resistance	Ω

Use it when:

A battery of cells. Series wins when $R \gg r$; parallel wins when $R \ll r$.

Trap:

In series the internal resistances add too — which is why stacking cells does not raise the current indefinitely.

● P3.11 Electrical power and energy

$$P = VI = I^2 R = \frac{V^2}{R} \quad W = VIt \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Symbol	Meaning	Unit
P	power dissipated	W
W	energy consumed	J or kWh
t	time	s

Use it when:

Heating, power ratings, or an electricity bill.

Trap:

Choose the right form. In **series** the current is shared so $P = I^2 R$ applies and the largest resistor dissipates most; in **parallel** the voltage is shared so $P = V^2/R$ applies and the *smallest* resistor dissipates most.

● P3.12 Kirchhoff's two rules

$$\text{junction: } \sum I = 0 \quad \text{loop: } \sum \Delta V = 0$$

Symbol	Meaning	Unit
junction rule	conservation of charge	A
loop rule	conservation of energy	V

Use it when:

A network that series and parallel reduction cannot simplify. Naming which conservation law each rule expresses is usually worth a mark on its own.

Trap:

Fix a sign convention before you start and keep it. Going through a resistor along the current gives $-IR$; entering a cell at its negative terminal gives $+\mathcal{E}$.

● P3.13 Wheatstone bridge, and the metre bridge

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \text{metre bridge: } X = \frac{R(100 - l)}{l}$$

Symbol	Meaning	Unit
X	unknown resistance	Ω
R	known resistance in the other gap	Ω
l	balancing length from the left end	cm

Use it when:

A galvanometer reads zero — the balance condition. At balance, no current flows through the galvanometer arm, so it can be removed entirely.

Trap:

The bridge is most sensitive when all four resistances are comparable, which is why the balance point should sit near the middle of the wire.

● P3.14 Potentiometer — comparing emfs and finding internal resistance

$$V = Kl, \quad K = \frac{V}{L} \quad \frac{\mathcal{E}_1}{\mathcal{E}_2} = \frac{l_1}{l_2} \quad r = R \left(\frac{l_1 - l_2}{l_2} \right)$$

Symbol	Meaning	Unit
K	potential gradient along the wire	V m^{-1} or V cm^{-1}
l_1	balancing length with the cell open	cm
l_2	balancing length with R across the cell	cm

Use it when:

Comparing two emfs, or measuring internal resistance.

Trap:

A potentiometer beats a voltmeter because at balance it draws **no current** from the cell, so it measures true emf rather than terminal voltage. Sensitivity improves with a **longer** wire — a smaller potential gradient.

CH 4 Moving Charges and Magnetism — 12 entries

Recognise strip

- P4.1 Biot-Savart law
- P4.2 Field at the centre of a circular loop
- P4.3 Field on the axis of a circular loop
- P4.4 Ampère's circuital law
- P4.5 Field of a long straight wire
- P4.6 Field inside a solenoid and a toroid
- P4.7 Force on a moving charge, and the Lorentz force
- P4.8 Radius and period of a charged particle's circular path
- P4.9 Force on a current-carrying conductor
- P4.10 Force between two parallel currents, and the ampere
- P4.11 Torque on a current loop, and magnetic moment
- P4.12 Galvanometer, and converting it to an ammeter or voltmeter

● P4.1 Biot-Savart law

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2} \quad d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Symbol	Meaning	Unit
dB	field from one current element	T
μ_0	permeability of free space	$4\pi \times 10^{-7} \text{ T m A}^{-1}$
dl	length of the current element	m
θ	angle between $d\vec{l}$ and $\hat{r}$	rad or $^\circ$

Use it when:

Deriving the field of any current shape. The magnetic analogue of Coulomb's law.

Trap:

$dB = 0$ straight ahead of the element ($\theta = 0$) and maximum sideways ($\theta = 90^\circ$) — unlike the electric field, which is maximum along the line.

● **P4.2 Field at the centre of a circular loop**

$$B = \frac{\mu_0 I}{2R} \quad \text{for } N \text{ turns: } B = \frac{\mu_0 NI}{2R}$$

Symbol	Meaning	Unit
B	magnetic field at the centre	T
R	radius of the loop	m
N	number of turns	dimensionless

Use it when:

A circular coil and the field at its centre. For an arc subtending angle ϕ radians, multiply by $\phi/2\pi$.

Trap:

Only for the **centre**. Off-axis needs P4.3, and R is the radius, not the diameter.

○ **P4.3 Field on the axis of a circular loop**

$$B = \frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}} \quad \text{for } N \text{ turns, multiply by } N$$

Symbol	Meaning	Unit
x	distance along the axis from the centre	m
R	loop radius	m

Use it when:

The point is on the axis but not at the centre. Setting $x = 0$ recovers P4.2 — use that as your check.

Trap:

Far away ($x \gg R$) this becomes $\mu_0 \cdot 2M/4\pi x^3$ with $M = NIA$ — the loop behaves as a magnetic dipole.

● P4.4 Ampère's circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Symbol	Meaning	Unit
I_{enc}	current threading the closed loop	A
$d\vec{l}$	element of the Amperian loop	m

Use it when:

Symmetry lets you choose a loop on which B is constant — the magnetic counterpart of Gauss's law.

Trap:

Only **enclosed** current counts, and only the component of $\vec{B}$ along the path contributes. In Chapter 8 this law gains a second term.

● P4.5 Field of a long straight wire

$$B = \frac{\mu_0 I}{2\pi r} \quad \text{finite wire: } B = \frac{\mu_0 I}{4\pi r} (\sin \alpha_1 + \sin \alpha_2)$$

Symbol	Meaning	Unit
r	perpendicular distance from the wire	m
α_1, α_2	angles subtended by the wire's two ends	rad or °

Use it when:

A straight current-carrying wire. Field lines are concentric circles; direction by the right-hand thumb rule.

Trap:

$1/r$, not $1/r^2$. The infinite-wire form is the finite one with both angles at 90° .

● P4.6 Field inside a solenoid and a toroid

$$B_{\text{solenoid}} = \mu_0 n I \quad B_{\text{toroid}} = \mu_0 n I, \quad n = \frac{N}{2\pi r}$$

Symbol	Meaning	Unit
n	turns per unit length	m^{-1}
N	total number of turns	dimensionless
r	mean radius of the toroid	m

Use it when:

A long solenoid or a toroid. The field is uniform inside and essentially zero outside.

Trap:

n is turns per **metre**, not the total. At the **end** of a solenoid the field is half the interior value.

● P4.7 Force on a moving charge, and the Lorentz force

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad \vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

Symbol	Meaning	Unit
$\vec{F}$	force on the charge	N
$\vec{v}$	velocity	m s^{-1}
$\vec{B}$	magnetic field	T

Use it when:

A charge moves through a magnetic field. In a velocity selector the two forces balance, giving $v = E/B$.

Trap:

The magnetic force is always perpendicular to $\vec{v}$, so it **does no work** and cannot change the particle's speed — only its direction. A charge at rest, or moving parallel to $\vec{B}$, feels no magnetic force at all.

● P4.8 Radius and period of a charged particle's circular path

$$r = \frac{mv}{qB} \quad T = \frac{2\pi m}{qB} \quad \omega = \frac{qB}{m}$$

$$\text{pitch of the helix } p = v \cos \theta \cdot \frac{2\pi m}{qB}$$

Symbol	Meaning	Unit
r	radius of the circular path	m
T	period of revolution	s
ω	cyclotron angular frequency	rad s ⁻¹

Use it when:

A charge enters a field perpendicular to it. If it enters at an angle, the path is a helix and the pitch formula applies.

Trap:

T and ω are **independent of speed and radius** — that is the whole principle of the cyclotron.

● P4.9 Force on a current-carrying conductor

$$\vec{F} = I (\vec{L} \times \vec{B}), \quad F = BIL \sin \theta$$

Symbol	Meaning	Unit
L	length of conductor in the field	m
I	current	A
θ	angle between the conductor and $\vec{B}$	rad or °

Use it when:

A wire in a magnetic field. Direction by Fleming's left-hand rule.

Trap:

Zero when the wire is **parallel** to the field, maximum when perpendicular. This is the same force as P4.7 summed over all the drifting charges.

● P4.10 Force between two parallel currents, and the ampere

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r} \quad \text{for } I_1 = I_2 = 1 \text{ A, } r = 1 \text{ m : } \frac{F}{L} = 2 \times 10^{-7} \text{ N m}^{-1}$$

Symbol	Meaning	Unit
F/L	force per unit length	N m ⁻¹
r	separation of the wires	m

Use it when:

Two parallel wires. This relation is the **definition of the ampere** — quoting the 2×10^{-7} case is the answer to "define the ampere".

Trap:

Currents in the **same** direction **attract**; opposite directions repel. That is the reverse of what charges do, and it is asked.

● **P4.11 Torque on a current loop, and magnetic moment**

$$\vec{\tau} = \vec{M} \times \vec{B}, \quad \tau = NIAB \sin \theta \quad M = NIA$$

Symbol	Meaning	Unit
M	magnetic dipole moment	A m ²
A	area of the loop	m ²
N	number of turns	dimensionless
θ	angle between $\vec{M}$ and $\vec{B}$	rad or °

Use it when:

A coil in a magnetic field — the working principle of the motor and the galvanometer.

Trap:

Torque is maximum when the coil's **plane** is parallel to $\vec{B}$ (so $\vec{M}$ is perpendicular to it), and zero when the plane is perpendicular. Net force is zero in a uniform field.

● **P4.12 Galvanometer, and converting it to an ammeter or voltmeter**

$$I = \frac{NBA}{k} \phi \quad \text{shunt: } S = \frac{I_g G}{I - I_g} \quad \text{series: } R = \frac{V}{I_g} - G$$

Symbol	Meaning	Unit
ϕ	deflection	rad or div
k	torsional constant of the suspension	N m rad ⁻¹
G	galvanometer resistance	Ω
I_g	current for full-scale deflection	A

Use it when:

Sensitivity, or converting the meter. **Current sensitivity** is NBA/k ; **voltage sensitivity** is NBA/kG .

Trap:

Raising current sensitivity does **not** automatically raise voltage sensitivity — adding turns raises N but also raises G . An ammeter needs a **small** shunt in parallel; a voltmeter needs a **large** resistance in series.

CH 5 Magnetism and Matter — 9 entries

Recognise strip

- P5.1 Gauss's law for magnetism
- P5.2 Magnetic moment of a bar magnet and of a coil
- P5.3 Field of a bar magnet, axial and equatorial
- P5.4 Torque, energy and work for a magnetic dipole
- P5.5 A magnet oscillating in a field
- P5.6 Earth's magnetic field and the angle of dip
- P5.7 Magnetisation, magnetising field and susceptibility
- P5.8 Permeability and its relation to susceptibility
- P5.9 Curie's law, and the three kinds of magnetic material

● P5.1 Gauss's law for magnetism

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

Symbol	Meaning	Unit
$\vec{B}$	magnetic field	T
$d\vec{S}$	element of any closed surface	m ²

Use it when:

Asked why magnetic monopoles do not exist, or to contrast with the electric Gauss's law.

Trap:

Always exactly zero, because magnetic field lines are **closed loops** — every line entering a surface leaves it. Cutting a magnet in half gives two magnets, never an isolated pole.

● P5.2 Magnetic moment of a bar magnet and of a coil

$$M = m \times 2l \quad M = NIA$$

Symbol	Meaning	Unit
M	magnetic dipole moment	A m ² or J T ⁻¹
m	pole strength	A m
$2l$	magnetic length	m
N, I, A	turns, current, area of a coil	—, A, m ²

Use it when:

Any dipole calculation. A current loop and a bar magnet are interchangeable once you know M .

Trap:

Magnetic length is about $\frac{5}{6}$ of the geometric length of a bar magnet. Direction runs **S to N** inside the magnet.

● **P5.3 Field of a bar magnet, axial and equatorial**

$$B_{\text{axial}} = \frac{\mu_0}{4\pi} \frac{2M}{r^3} \quad B_{\text{equatorial}} = \frac{\mu_0}{4\pi} \frac{M}{r^3}$$

Symbol	Meaning	Unit
B	field at distance r	T
M	magnetic moment	A m ²
r	distance from the centre, $r \gg l$	m

Use it when:

A short bar magnet, far from it. Identical in form to the electric dipole (P1.8, P1.9) with $\mu_0/4\pi$ replacing $1/4\pi\epsilon_0$.

Trap:

Axial is **twice** equatorial, and the two point in opposite senses. The same factor of 2 as in electrostatics.

● **P5.4 Torque, energy and work for a magnetic dipole**

$$\vec{\tau} = \vec{M} \times \vec{B}, \quad \tau = MB \sin \theta \quad U = -\vec{M} \cdot \vec{B}$$

$$W = MB (\cos \theta_1 - \cos \theta_2)$$

Symbol	Meaning	Unit
τ	torque on the magnet	N m
U	potential energy	J
W	work to turn from θ_1 to θ_2	J

Use it when:

A magnet is rotated in a field. Turning from aligned to fully reversed costs $W = 2MB$.

Trap:

Structurally identical to the electric dipole (P1.10). $\theta = 0$ stable, $\theta = 180^\circ$ unstable.

○ P5.5 A magnet oscillating in a field

$$T = 2\pi\sqrt{\frac{I}{MB}} \quad B = \frac{4\pi^2 I}{MT^2}$$

Symbol	Meaning	Unit
T	period of oscillation	s
I	moment of inertia , not current	kg m ²
M	magnetic moment	A m ²

Use it when:

A magnet is displaced slightly and released — a vibration magnetometer. Small angles only, so that $\sin \theta \approx \theta$.

Trap:

I here is moment of inertia. For a bar of mass m and length L about its centre, $I = mL^2/12$.

○ P5.6 Earth's magnetic field and the angle of dip

$$B_H = B \cos \delta, \quad B_V = B \sin \delta \quad \tan \delta = \frac{B_V}{B_H}, \quad B = \sqrt{B_H^2 + B_V^2}$$

Symbol	Meaning	Unit
B_H	horizontal component	T
B_V	vertical component	T
δ	angle of dip (inclination)	°

Use it when:

Earth's field is resolved. The three elements are declination, dip and horizontal component.

Trap:

$\delta = 0$ at the magnetic equator and 90° at the poles. At the equator the field is entirely horizontal, so a dip needle rests flat.

○ P5.7 Magnetisation, magnetising field and susceptibility

$$I = \frac{M}{V} \quad \chi_m = \frac{I}{H} \quad B = \mu_0 (H + I) = \mu_0 H (1 + \chi_m)$$

Symbol	Meaning	Unit
I	intensity of magnetisation	A m^{-1}
H	magnetising field intensity	A m^{-1}
χ_m	magnetic susceptibility	dimensionless

Use it when:

A material is placed in a field. χ_m is small and negative for diamagnetics, small and positive for paramagnetics, and very large for ferromagnetics.

Trap:

This I is magnetisation — the third meaning of the letter in this chapter, after current and moment of inertia. H and I share a unit; B does not.

○ P5.8 Permeability and its relation to susceptibility

$$\mu = \mu_0 (1 + \chi_m) \quad \mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m$$

Symbol	Meaning	Unit
μ	absolute permeability	T m A^{-1}
μ_r	relative permeability	dimensionless
μ_0	permeability of free space	$4\pi \times 10^{-7} \text{ T m A}^{-1}$

Use it when:

Converting between susceptibility and permeability. $\mu_r < 1$ diamagnetic, slightly > 1 paramagnetic, $\gg 1$ ferromagnetic.

Trap:

χ_m is dimensionless but μ is not — and μ_r is the one that has no unit.

● P5.9 Curie's law, and the three kinds of magnetic material

$$\chi_m = \frac{C}{T} \quad (\text{paramagnetic; ferromagnetic above } T_C : \chi_m = \frac{C}{T - T_C})$$

Symbol	Meaning	Unit
C	Curie constant	K
T	absolute temperature	K
T_C	Curie temperature	K

Use it when:

Susceptibility against temperature. Heating randomises the aligned dipoles, so χ_m falls.

Trap:

Above T_C a ferromagnet becomes **paramagnetic**, not diamagnetic. Diamagnetism alone is **temperature-independent** — Curie's law does not apply to it.

CH 6 Electromagnetic Induction — 9 entries

Recognise strip

- P6.1 Magnetic flux
- P6.2 Faraday's law and Lenz's law
- P6.3 Motional emf from a rod moving in a field
- P6.4 A rod rotating about one end
- P6.5 Charge that flows during a flux change
- P6.6 Self-inductance, and that of a solenoid
- P6.7 Mutual inductance of two coaxial solenoids
- P6.8 Energy stored in an inductor, and magnetic energy density
- P6.9 The AC generator

● P6.1 Magnetic flux

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Symbol	Meaning	Unit
Φ_B	magnetic flux	Wb (weber)
A	area of the loop	m ²
θ	angle between $\vec{B}$ and the area normal	rad or °

Use it when:

Anything about induction — flux is the quantity whose change drives everything in this chapter.

Trap:

θ is measured from the **normal** to the plane, not from the plane itself. A coil lying flat in a vertical field has $\theta = 0$ and maximum flux, not zero.

● P6.2 Faraday's law and Lenz's law

$$\varepsilon = -N \frac{d\Phi_B}{dt} \quad I = \frac{\varepsilon}{R} = -\frac{N}{R} \frac{d\Phi_B}{dt}$$

Symbol	Meaning	Unit
ε	induced emf	V
N	number of turns	dimensionless
$d\Phi_B/dt$	rate of change of flux	Wb s ⁻¹

Use it when:

Flux changes for any reason — B changing, A changing, or the coil rotating.

Trap:

The minus sign **is** Lenz's law: the induced current opposes the change that produced it. It is a consequence of **conservation of energy** — that is the answer when asked to justify it.

● **P6.3 Motional emf from a rod moving in a field**

$$\varepsilon = Blv \quad I = \frac{Blv}{R}, \quad F = \frac{B^2 l^2 v}{R}, \quad P = \frac{B^2 l^2 v^2}{R}$$

Symbol	Meaning	Unit
l	length of the rod in the field	m
v	speed of the rod	m s ⁻¹
F	opposing force needed to keep it moving	N

Use it when:

A conductor slides on rails. B , l and v must be mutually perpendicular.

Trap:

Power applied equals power dissipated — the mechanical work done against the opposing force **is** the electrical energy produced. That equality is a favourite question.

○ **P6.4 A rod rotating about one end**

$$\varepsilon = \frac{1}{2} B \omega l^2$$

Symbol	Meaning	Unit
ω	angular velocity	rad s ⁻¹
l	length of the rod	m
ε	emf between centre and rim	V

Use it when:

A rod or disc spins in a perpendicular field.

Trap:

The $\frac{1}{2}$ comes from the average speed along the rod — the far end moves fastest, the pivot not at all. $\omega = 2\pi f$, so a rod at 50 rev/s has $\omega = 100\pi$.

○ **P6.5 Charge that flows during a flux change**

$$q = \frac{N \Delta \Phi_B}{R}$$

Symbol	Meaning	Unit
q	total charge circulated	C
$\Delta \Phi_B$	total change in flux	Wb
R	total circuit resistance	Ω

Use it when:

Asked for charge rather than current — a magnet dropped through a coil, or a coil flipped over.

Trap:

Independent of **how fast** the change happens. Flipping a coil through 180° changes flux by $2BA$, not BA .

● **P6.6 Self-inductance, and that of a solenoid**

$$N\Phi = LI \quad \varepsilon = -L \frac{dI}{dt} \quad L = \mu_0 n^2 A l = \frac{\mu_0 N^2 A}{l}$$

Symbol	Meaning	Unit
L	self-inductance	H (henry)
l	length of the solenoid	m
n	turns per unit length, N/l	m^{-1}
A	cross-sectional area	m^2

Use it when:

A coil opposes a change in its own current — electrical inertia. A core of relative permeability μ_r multiplies L by μ_r .

Trap:

Here l is **length** and L is **inductance** — the two are easy to collide. L goes as N^2 , so doubling the turns quadruples the inductance.

○ P6.7 Mutual inductance of two coaxial solenoids

$$\varepsilon_2 = -M \frac{dI_1}{dt} \quad M = \frac{\mu_0 N_1 N_2 A}{l} \quad M_{12} = M_{21}$$

Symbol	Meaning	Unit
M	mutual inductance	H
N_1, N_2	turns on each solenoid	dimensionless
A	area of the inner solenoid	m ²

Use it when:

Two coupled coils — the basis of the transformer.

Trap:

Use the area of the **inner** coil, since that is all the flux the outer one links. $M_{12} = M_{21}$ always, however different the two coils are.

● P6.8 Energy stored in an inductor, and magnetic energy density

$$U = \frac{1}{2}LI^2 \quad u = \frac{B^2}{2\mu_0}$$

Symbol	Meaning	Unit
U	energy stored in the magnetic field	J
u	energy per unit volume	J m ⁻³
B	field inside the inductor	T

Use it when:

Energy in a coil. Exactly parallel to the capacitor: $\frac{1}{2}CV^2 \leftrightarrow \frac{1}{2}LI^2$, and $\frac{1}{2}\varepsilon_0 E^2 \leftrightarrow B^2/2\mu_0$.

Trap:

Note μ_0 is in the **denominator** for magnetic energy density, where ε_0 is in the numerator for electric.

● P6.9 The AC generator

$$\varepsilon = NAB\omega \sin(\omega t) = \varepsilon_0 \sin(\omega t), \quad \varepsilon_0 = NAB\omega$$

Symbol	Meaning	Unit
ε_0	peak emf	V
ω	angular frequency of rotation, $2\pi f$	rad s ⁻¹
N, A, B	turns, coil area, field	—, m ² , T

Use it when:

A coil rotates in a uniform field. Converts mechanical energy into electrical.

Trap:

Emf is **maximum** when the coil's plane is *parallel* to $\vec{B}$ (flux momentarily zero but changing fastest), and zero when the plane is perpendicular. The intuition runs backwards from most students' first guess.

CH 7 Alternating Current — 11 entries

Recognise strip

- P7.1 RMS and mean values of an alternating quantity
- P7.2 AC through a pure resistor
- P7.3 AC through a pure inductor
- P7.4 AC through a pure capacitor
- P7.5 Series LCR — impedance and phase angle
- P7.6 Resonance in a series LCR circuit
- P7.7 Sharpness of resonance — the Q factor
- P7.8 Average power, power factor and wattless current
- P7.9 LC oscillations
- P7.10 The transformer
- P7.11 Why transformers lose energy

● P7.1 RMS and mean values of an alternating quantity

$$I_{\text{rms}} = \frac{i_0}{\sqrt{2}} \approx 0.707 i_0 \quad I_{\text{mean}} = \frac{2i_0}{\pi} \approx 0.637 i_0$$

Symbol	Meaning	Unit
i_0	peak current	A
I_{rms}	root-mean-square (virtual) current	A
I_{mean}	mean over a half cycle	A

Use it when:

Converting between peak and stated values. Mains "220 V" is the rms value, so the peak is about 311 V.

Trap:

The mean over a **full** cycle is **zero** — which is exactly why rms exists. Every ammeter and

voltmeter reads rms.

● P7.2 AC through a pure resistor

$$i = i_0 \sin \omega t, \quad i_0 = \frac{e_0}{R}, \quad \phi = 0$$

Symbol	Meaning	Unit
ϕ	phase difference between V and I	rad
R	resistance	Ω

Use it when:

Only a resistor is present. Current and voltage are **in phase**.

Trap:

Resistance does not depend on frequency, unlike both reactances.

● P7.3 AC through a pure inductor

$$X_L = \omega L = 2\pi f L \quad i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right), \quad i_0 = \frac{e_0}{X_L}$$

Symbol	Meaning	Unit
X_L	inductive reactance	Ω
L	inductance	H
f	frequency	Hz

Use it when:

A coil in an AC circuit. Current **lags** voltage by 90° .

Trap:

$X_L \propto f$, so an inductor blocks high frequencies and passes DC ($f = 0$, $X_L = 0$) freely. It is a choke.

● P7.4 AC through a pure capacitor

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \quad i = i_0 \sin \left(\omega t + \frac{\pi}{2} \right), \quad i_0 = \frac{e_0}{X_C}$$

Symbol	Meaning	Unit
X_C	capacitive reactance	Ω
C	capacitance	F

Use it when:

A capacitor in an AC circuit. Current **leads** voltage by 90° .

Trap:

$X_C \propto 1/f$, the opposite of the inductor: a capacitor blocks DC completely ($f = 0$, $X_C \rightarrow \infty$) and passes high frequencies. Remember the order as **CIVIL** — in C, I leads V; V leads I in L.

● P7.5 Series LCR — impedance and phase angle

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \tan \phi = \frac{X_L - X_C}{R} \quad E = IZ$$

Symbol	Meaning	Unit
Z	impedance	Ω
ϕ	phase angle between E and I	rad or $^\circ$
X_L, X_C	the two reactances	Ω

Use it when:

R, L and C in series. Found from the phasor diagram, where V_L and V_C are antiparallel so they subtract.

Trap:

$X_L > X_C \rightarrow$ inductive, current lags. $X_L < X_C \rightarrow$ capacitive, current leads. Because V_L and V_C oppose, the voltage across one of them can **exceed the supply voltage** — which is not an error.

● P7.6 Resonance in a series LCR circuit

$$X_L = X_C \Rightarrow \omega_r = \frac{1}{\sqrt{LC}}, \quad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{at resonance: } Z = R \text{ (minimum), } I = \frac{E}{R} \text{ (maximum), } \phi = 0$$

Symbol	Meaning	Unit
ω_r	resonant angular frequency	rad s ⁻¹
f_r	resonant frequency	Hz

Use it when:

The circuit is tuned — a radio receiver selecting a station.

Trap:

At resonance the circuit is purely **resistive** and power factor is 1. Resonance only exists in a series circuit if both L and C are present — an RL or RC circuit never resonates.

○ P7.7 Sharpness of resonance — the Q factor

$$Q = \frac{\omega_r L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_r}{\Delta\omega}$$

Symbol	Meaning	Unit
Q	quality factor	dimensionless
$\Delta\omega$	bandwidth	rad s ⁻¹
R	resistance in the circuit	Ω

Use it when:

Asked how sharply the circuit is tuned. High Q means a narrow, selective peak.

Trap:

Q rises as R **falls** — a low-resistance circuit is the sharply tuned one.

● P7.8 Average power, power factor and wattless current

$$P_{\text{avg}} = E_{\text{rms}} I_{\text{rms}} \cos \phi \quad \text{power factor} = \cos \phi = \frac{R}{Z}$$

Symbol	Meaning	Unit
P_{avg}	average power consumed	W
$\cos \phi$	power factor	dimensionless
Z	impedance	Ω

Use it when:

Power in any AC circuit. Only the resistance dissipates energy.

Trap:

In a **pure** inductor or capacitor $\phi = 90^\circ$, so $\cos \phi = 0$ and the average power is **zero** — the current is called **wattless**. Energy is stored and returned each quarter cycle, never consumed.

○ P7.9 LC oscillations

$$\frac{d^2 q}{dt^2} + \frac{q}{LC} = 0 \quad \omega = \frac{1}{\sqrt{LC}} \quad U = \frac{q^2}{2C} + \frac{1}{2} Li^2 = \text{constant}$$

Symbol	Meaning	Unit
q	charge on the capacitor	C
ω	angular frequency of oscillation	rad s ⁻¹
U	total energy, constant if $R = 0$	J

Use it when:

A charged capacitor is connected across an inductor. Energy sloshes between the electric field of C and the magnetic field of L, exactly like a mass on a spring.

Trap:

Undamped only in the idealised $R = 0$ case. Any real circuit has resistance, so the oscillations decay.

● P7.10 The transformer

$$\frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} \quad \text{ideal: } \varepsilon_p I_p = \varepsilon_s I_s$$

Symbol	Meaning	Unit
N_p, N_s	turns on primary and secondary	dimensionless
$\varepsilon_p, \varepsilon_s$	primary and secondary voltages	V
I_p, I_s	primary and secondary currents	A

Use it when:

Voltage is stepped up or down. Step-up means more secondary turns — and correspondingly **less** secondary current.

Trap:

A transformer works on **AC only** — DC produces no changing flux, so no induced emf. It never creates energy: what it gains in voltage it loses in current.

○ P7.11 Why transformers lose energy

$$\eta = \frac{\text{output power}}{\text{input power}} \times 100\%$$

Symbol	Meaning	Unit
flux leakage	not all primary flux links the secondary	—
copper loss	$I^2 R$ heating in the windings	W
eddy currents	induced loops in the core — reduced by laminating it	W
hysteresis	repeated remagnetisation of the core	W

Use it when:

"State four energy losses in a transformer and how each is minimised" — a standing question with four recallable answers.

Trap:

Each loss has its own remedy: thick copper wire for copper loss, a laminated core for eddy currents, a soft-iron core for hysteresis, and winding one coil over the other for flux leakage.

CH 8 Electromagnetic Waves — 5 entries

Recognise strip

- P8.1 Displacement current
- P8.2 Ampère–Maxwell law
- P8.3 Speed of an electromagnetic wave
- P8.4 The wave itself — fields, energy and momentum
- P8.5 The electromagnetic spectrum, in order

● P8.1 Displacement current

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Symbol	Meaning	Unit
I_d	displacement current	A
Φ_E	electric flux	V m
ϵ_0	permittivity of free space	$8.854 \times 10^{-12} \text{ F m}^{-1}$

Use it when:

A charging capacitor. It exists in the **gap between the plates**, where no charge flows, and is exactly equal to the conduction current in the wires.

Trap:

It is not a flow of charge. Maxwell introduced it to fix an inconsistency in Ampère's law: two surfaces bounded by the same loop gave different answers, one passing through the wire and one through the gap.

● P8.2 Ampère–Maxwell law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_c + \epsilon_0 \frac{d\Phi_E}{dt} \right) = \mu_0 (I_c + I_d)$$

Symbol	Meaning	Unit
I_c	conduction current	A
I_d	displacement current	A

Use it when:

Asked for the modified Ampère's law or to state Maxwell's equations. This is P4.4 with the new term added.

Trap:

The consequence is the whole chapter: a changing electric field produces a magnetic field, and a changing magnetic field produces an electric field — so the two sustain each other and propagate as a wave.

● P8.3 Speed of an electromagnetic wave

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1} \quad v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{n}$$

Symbol	Meaning	Unit
c	speed in vacuum	m s^{-1}
v	speed in a medium	m s^{-1}
n	refractive index of the medium	dimensionless

Use it when:

Asked why light is an electromagnetic wave — Maxwell's predicted speed matched the measured speed of light, which is what identified them.

Trap:

Built from two constants measured in purely electric and magnetic experiments, with no light involved. That is what made the result so striking.

● P8.4 The wave itself — fields, energy and momentum

$$E_y = E_0 \sin(\omega t - kx), \quad B_z = B_0 \sin(\omega t - kx), \quad k = \frac{2\pi}{\lambda}$$

$$c = \frac{E_0}{B_0} \quad p = \frac{U}{c} \quad u_{\text{avg}} = \frac{1}{2} \epsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}$$

Symbol	Meaning	Unit
E_0, B_0	peak field amplitudes	$\text{V m}^{-1}, \text{T}$
k	wave number	m^{-1}
p	momentum delivered	kg m s^{-1}
U	energy delivered	J

Use it when:

Given one field amplitude and asked for the other, or asked about radiation pressure.

Trap:

$\vec{E}$, $\vec{B}$ and the direction of propagation are **mutually perpendicular**, in that order, and the two fields are **in phase**. Energy is shared equally between them. For a *totally reflecting* surface the momentum delivered is $2U/c$.

● P8.5 The electromagnetic spectrum, in order

$$c = f\lambda \quad E = hf$$

Symbol	Meaning	Unit
order	radio → microwave → infrared → visible → UV → X-ray → γ-ray	—
λ	decreases along that order	m
f, E	increase along that order	Hz, J
visible	about 400 nm (violet) to 700 nm (red)	nm

Use it when:

Ordering by wavelength or frequency, or naming a source and use. Radio from oscillating circuits; microwaves from klystrons, used in radar and ovens; infrared from hot bodies, used in therapy and remote controls; UV from the sun and arcs; X-rays from decelerating electrons; γ-rays from nuclei.

Trap:

All travel at the **same speed** c in vacuum. Only wavelength and frequency differ, and they always move in opposite directions along the list.

CH 9 Ray Optics and Optical Instruments — 12 entries

Written from NCERT and the published Ray Optics to 9.4 page. The lecture transcripts for this chapter are not yet processed, so the emphasis here follows the textbook rather than the teacher; it will be revisited once those are in.

Recognise strip

- P9.1 Mirror formula and magnification

- P9.2 Focal length from radius of curvature
- P9.3 Snell's law and refractive index
- P9.4 Real depth, apparent depth and the shift
- P9.5 Critical angle and total internal reflection
- P9.6 Refraction at a single spherical surface
- P9.7 Lens maker's formula
- P9.8 Thin lens formula and magnification
- P9.9 Power of a lens, and lenses in contact
- P9.10 Refraction through a prism
- P9.11 Magnifying power of a simple and compound microscope
- P9.12 Magnifying power of a telescope

● P9.1 Mirror formula and magnification

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad m = \frac{h'}{h} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

Symbol	Meaning	Unit
u	object distance, from the pole	m
v	image distance	m
f	focal length	m
m	linear magnification	dimensionless

Use it when:

Any spherical mirror. All distances are measured from the **pole** under the New Cartesian convention.

Trap:

Note the **plus** sign between $1/v$ and $1/u$ — the lens formula has a minus. Negative m means real and inverted; concave f is negative, convex f positive.

● P9.2 Focal length from radius of curvature

$$f = \frac{R}{2}$$

Symbol	Meaning	Unit
R	radius of curvature	m
f	focal length	m

Use it when:

A mirror is described by its radius rather than its focal length.

Trap:

Holds for **small aperture** only, and for mirrors — not for lenses, where the lens maker's formula applies instead.

● **P9.3 Snell's law and refractive index**

$$\frac{\sin i}{\sin r} = {}_1n_2 = \frac{n_2}{n_1} \iff n_1 \sin i = n_2 \sin r$$

$$n = \frac{c}{v} = \frac{\lambda_{\text{air}}}{\lambda_{\text{medium}}} \quad {}_1n_2 = \frac{1}{{}_2n_1}$$

Symbol	Meaning	Unit
i, r	angles of incidence and refraction	°
n	refractive index	dimensionless
v	speed of light in the medium	m s ⁻¹

Use it when:

Light crosses a boundary. Some papers write μ instead of n — same quantity.

Trap:

Frequency does not change on refraction — it is fixed by the source. Since $v = f\lambda$, it is the wavelength that shortens in the denser medium.

● **P9.4 Real depth, apparent depth and the shift**

$$n = \frac{\text{real depth}}{\text{apparent depth}} \quad \text{shift } x = t \left(1 - \frac{1}{n} \right)$$

Symbol	Meaning	Unit
t	real depth or slab thickness	m
x	apparent shift	m
n	refractive index of the denser medium	dimensionless

Use it when:

A coin in water, a pin under a glass slab, a pool that looks shallower than it is.

Trap:

Valid for **near-normal viewing** only. The shift does not depend on where the slab sits between object and eye.

● P9.5 Critical angle and total internal reflection

$$\sin C = \frac{n_{\text{rarer}}}{n_{\text{denser}}} = \frac{1}{n} \quad C = \sin^{-1} \left(\frac{1}{n} \right)$$

Symbol	Meaning	Unit
C	critical angle	°
n	index of the denser medium w.r.t. the rarer	dimensionless

Use it when:

Optical fibres, mirages, the brilliance of diamond ($C \approx 24^\circ$), totally reflecting prisms.

Trap:

Two conditions, both required: light must travel **denser to rarer**, and i must **exceed** C . Since n is larger for violet, violet has the smallest C and is totally reflected first.

○ P9.6 Refraction at a single spherical surface

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Symbol	Meaning	Unit
n_1	index of the medium the light starts in	dimensionless
n_2	index of the medium it enters	dimensionless
R	radius of curvature of the surface	m

Use it when:

One curved refracting surface. Applying it twice, once at each face, is what produces the lens maker's formula.

Trap:

Distances are measured from the **pole of the surface**, and the sign convention still applies to R .

● P9.7 Lens maker's formula

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Symbol	Meaning	Unit
n	index of the lens w.r.t. the surrounding medium	dimensionless
R_1	radius of the first surface met	m
R_2	radius of the second surface	m

Use it when:

Focal length from the lens's shape and material, or explaining what happens when a lens is moved into water.

Trap:

It is n **relative to the surroundings**. A glass lens ($n = 1.5$) in water ($n = 1.33$) has a much longer focal length; if the surrounding medium had the same index as the lens, f would be infinite and the lens would vanish optically.

● P9.8 Thin lens formula and magnification

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad m = \frac{h'}{h} = \frac{v}{u}$$

Symbol	Meaning	Unit
u, v	object and image distances from the optical centre	m
f	focal length — positive convex, negative concave	m
m	magnification	dimensionless

Use it when:

Any thin lens.

Trap:

Minus for lenses, **plus** for mirrors — and the magnification is $+v/u$ for a lens but $-v/u$ for a mirror. Both differences are exam-critical.

● P9.9 Power of a lens, and lenses in contact

$$P = \frac{1}{f \text{ in metres}} \quad P = P_1 + P_2 + \dots \quad \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \dots$$

Symbol	Meaning	Unit
P	power of the lens	D (diopetre)
f	focal length — must be in metres	m
F	focal length of the combination	m

Use it when:

Spectacle prescriptions, or two lenses stuck together.

Trap:

f in **metres**, not centimetres — a 20 cm lens is 5 D, not 0.05 D. Converging power is positive, diverging negative.

● P9.10 Refraction through a prism

$$A + \delta = i + e \quad r_1 + r_2 = A \quad n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Symbol	Meaning	Unit
A	angle of the prism	°
δ	angle of deviation	°
δ_m	minimum deviation	°
i, e	angles of incidence and emergence	°

Use it when:

A prism. At minimum deviation the ray passes **symmetrically**: $i = e$ and $r_1 = r_2 = A/2$.

Trap:

For a **thin** prism the formula simplifies to $\delta = (n - 1)A$. The graph of δ against i is a curve with a single minimum, not a straight line.

● P9.11 Magnifying power of a simple and compound microscope

$$\text{simple: } m = 1 + \frac{D}{f} \text{ (image at } D) \quad m = \frac{D}{f} \text{ (image at } \infty)$$

$$\text{compound: } m = \frac{v_o}{u_o} \left(1 + \frac{D}{f_e}\right) \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$$

Symbol	Meaning	Unit
D	least distance of distinct vision	25 cm
f_o, f_e	focal lengths of objective and eyepiece	m or cm
L	tube length (separation of the lenses)	m or cm

Use it when:

A magnifier or a microscope. High magnification needs **both** focal lengths short.

Trap:

Two cases for every instrument — image at D (relaxed formula has the $1+$) and image at infinity (no $1+$). Read which the question wants.

● P9.12 Magnifying power of a telescope

$$m = \frac{f_o}{f_e} \text{ (normal adjustment), } L = f_o + f_e \quad m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right) \text{ (image at } D\text{)}$$

Symbol	Meaning	Unit
f_o	objective focal length — long	m or cm
f_e	eyepiece focal length — short	m or cm
L	length of the telescope	m or cm

Use it when:

An astronomical telescope. A large objective also gathers more light, improving resolution and brightness.

Trap:

Exactly the **opposite** requirement from a microscope: the telescope objective wants a **long** focal length, the microscope objective a short one. A reflecting telescope replaces the objective lens with a concave mirror, avoiding chromatic aberration entirely.

CONST Constants and conversions — *know the units too*

● K Every constant this paper can hand you, with its unit

Symbol	Meaning	Value and unit
e	elementary charge	$1.6 \times 10^{-19} \text{ C}$
m_e	electron mass	$9.1 \times 10^{-31} \text{ kg}$
ϵ_0	permittivity of free space	$8.854 \times 10^{-12} \text{ F m}^{-1}$
$1/4\pi\epsilon_0$	Coulomb constant	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
μ_0	permeability of free space	$4\pi \times 10^{-7} \text{ T m A}^{-1}$
$\mu_0/4\pi$	the form used in Biot-Savart	$10^{-7} \text{ T m A}^{-1}$
c	speed of light in vacuum	$3 \times 10^8 \text{ m s}^{-1}$
h	Planck constant	$6.63 \times 10^{-34} \text{ J s}$
k_B	Boltzmann constant	$1.38 \times 10^{-23} \text{ J K}^{-1}$
μ_B	Bohr magneton	$9.27 \times 10^{-24} \text{ A m}^2$
1 eV	electron-volt	$1.6 \times 10^{-19} \text{ J}$
1 kWh	unit of electrical energy	$3.6 \times 10^6 \text{ J}$
D	least distance of distinct vision	25 cm
B_E	Earth's magnetic field, order of magnitude	$\sim 10^{-5} \text{ T}$

Trap:

Watch the letters that mean different things in different chapters: L is length (Ch 2, 4), self-inductance (Ch 6) and tube length (Ch 9); I is current, moment of inertia (Ch 5) and magnetisation (Ch 5); σ is surface charge density (Ch 1) and conductivity (Ch 3); μ is mobility (Ch 3) and permeability (Ch 5).

Built from the notes for Chapters 1–8 in this repository, which were themselves grounded against the lecture board frames rather than the ASR transcripts — the extracted NCERT text flattens equations during PDF conversion and is unreliable for them, so NCERT was used here only for units, constants and symbol names. Chapter 9 is written from NCERT and from the published **Ray Optics to 9.4** page, pending processing of that chapter's eighteen lecture videos.

Derivations for the entries marked ○, and for many marked ●, are on the companion page **Physics, Derived**.